

JAK / TYK2: 3 molecules stopped early, all small molecules

4 stopped trials, but 3 molecules — all small molecules: Baricitinib, Cravacitinib and Upadacitinib. 2 of the stops were the trial's own verdict and 1 followed a decision taken elsewhere; 1 cannot be established. Against 3.5% for all immunology & autoimmune trials with a resolved drug target and 3.3% across all immunology & autoimmune Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

5.1%

4 of 79 closed trials (95% CI 2.0%–12.3%)

2 / 0 / 0

efficacy / safety / both (adds to 4)

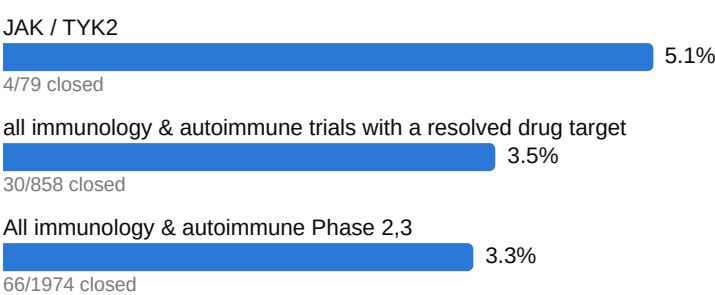
62.2%

of 127 trials have closed · 48 still open or unresolved

2.0%

stopped for a biological reason within 3 years of starting (95% CI 0.5%–7.9%; 47 still at risk; 1.9% area-wide)

How this compares



What is in this class: 37 distinct experimental drugs across 127 trials (small molecule 127, unknown 3, antibody 2, protein 1). Most tested: FILGOTINIB (25), BARICITINIB (20), TOFACITINIB (19), UPADACITINIB (19), CRAVACITINIB (18). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 4 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Eli Lilly) contributed 2; removing that programme's trials from both sides leaves 2.7%. Screened alongside 28 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 5 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.4%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	0	36	0.0%	0.0%–9.6%
2018–2020	4	35	11.4%	4.5%–26.0%
2021–2024	0	8	0.0%	0.0%–32.4%

Where the stops sit

Efficacy / futility: 2, Benefit–risk: 2.
Sponsors with most stops: Eli Lilly (2), Bristol Myers Squibb (1), AbbVie (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03599622	2	Bristol Myers Squibb BMJ	CRAVACITINIB	2022-12	Efficacy	Trial terminated because of lack of efficacy in the short term acute phase
NCT03843125	3	Eli Lilly LLY	BARICITINIB	2022-04	Benefit–risk	Study terminated due to insufficient evidence to support a positive benefit: risk profile. Safety findings were consistent with previously published OLUMIANT data
NCT03616912	3	Eli Lilly LLY	BARICITINIB	2021-11	Benefit–risk	Study terminated due to insufficient evidence to support a positive benefit: risk profile in systemic lupus erythematosus patients.
NCT03823378	2	AbbVie ABBV	ELSUBRUTINIB, UPADACITINIB	2020-09	Efficacy	Study M16-763 was terminated early as the benefit of each treatment arm from the feeder study (Study M16-063) did not provide appreciable evidence of differentiated clinical effect to warrant further long-term continuation.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 immunology & autoimmune trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 48% of industry immunology & autoimmune trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 62.2% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

Any mechanism, sponsor or indication, updated weekly. The dataset behind this brief covers every stopped immunology & autoimmune trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: clinicaltrialfailures.com/pricing

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.